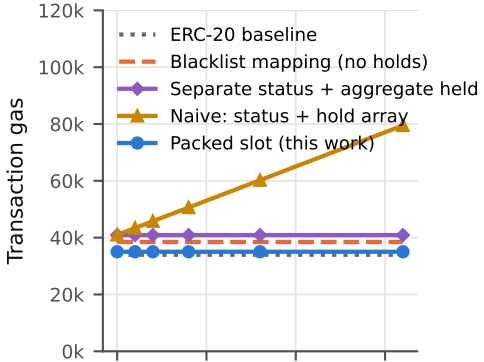
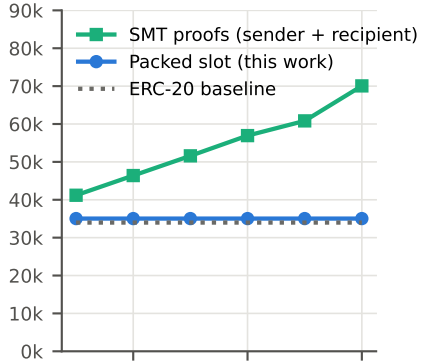
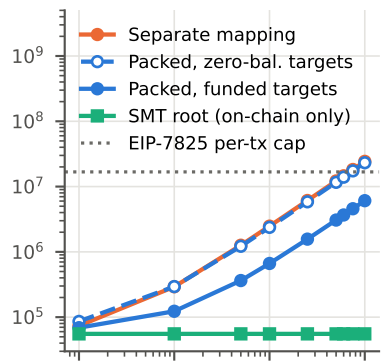

(a) Transfer cost vs. holds
Transaction gas
120k
100k
80k
60k
40k
20k
0k
0
5
10
15
Active holds on sender k
ERC-20 baseline
Blacklist mapping (no holds)
Separate status + aggregate held
Naive: status + hold array
Packed slot (this work)


(b) SMT transfer cost vs. list size
90k
80k
70k
60k
50k
40k
30k
20k
10k
0k
10^1
10^3
10^5
Restricted addresses N
SMT proofs (sender + recipient)
Packed slot (this work)
ERC-20 baseline


(c) List update cost (2-of-3)
10^9
10^8
10^7
10^6
10^5
10^0
10^1
10^2
10^3
Batch size K
Separate mapping
Packed, zero-bal. targets
Packed, funded targets
SMT root (on-chain only)
EIP-7825 per-tx cap